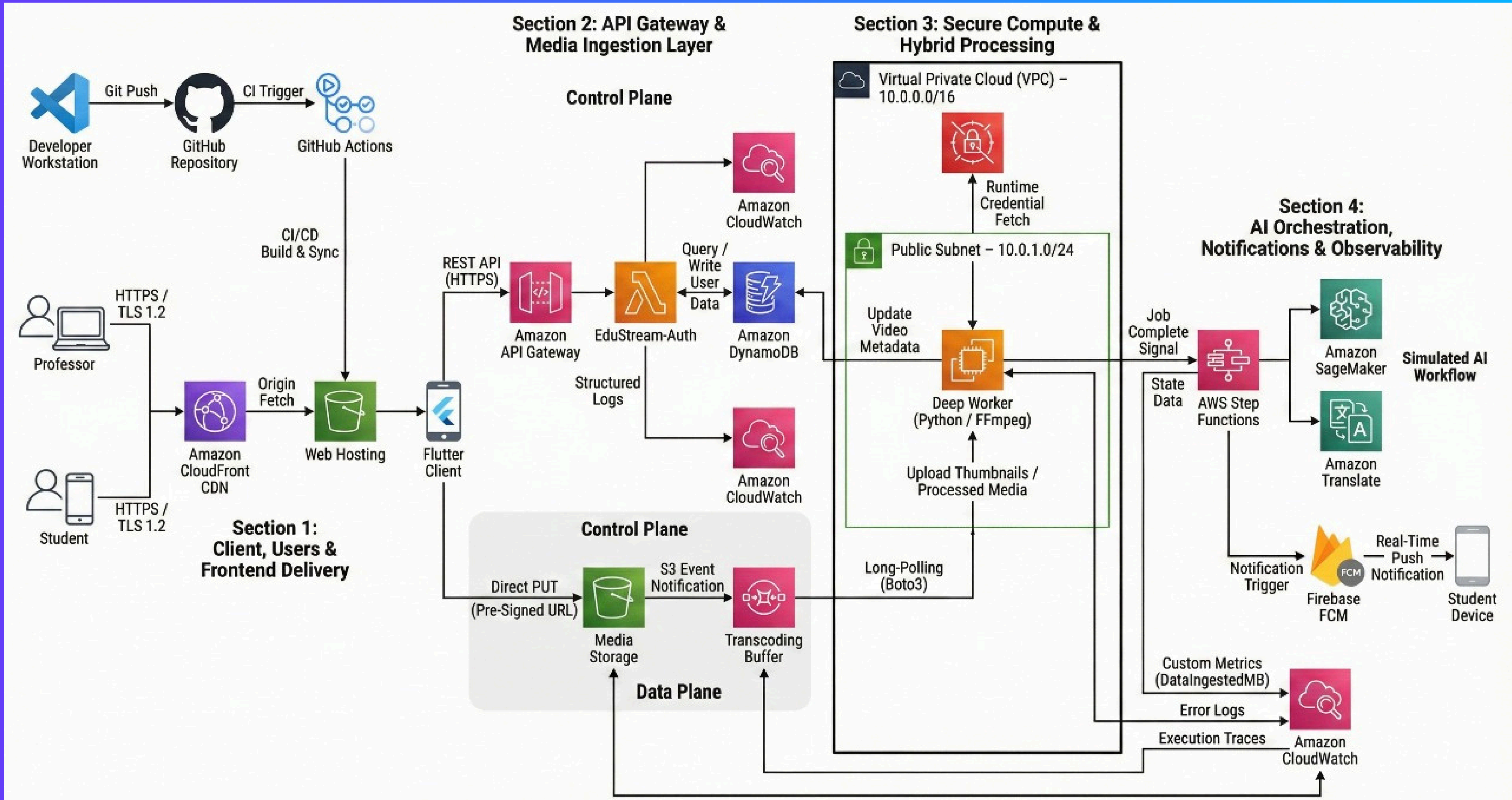




INTRODUCTION

EduStream replaces monolithic LMS with a Serverless/EC2 hybrid. By using asynchronous queues, we eliminate crashes and ensure 99.9% availability for heavy video workloads.

PROBLEM STATEMENT

- **Latency Bottlenecks:** Synchronous video processing blocks the main thread, causing App freezes.
- **Scalability Crises:** Monolithic backends crash during exam submissions.
- **Security Risks:** Hardcoded API keys in source code allow data breaches.

METHODOLOGY

- **Framework:** AWS Well-Architected compliant.
- **Decoupling:** SQS-buffered jobs (Netflix Pattern) to prevent overload.
- **Hybrid Compute:** Lambda for APIs; EC2 for FFmpeg transcoding.
- **IaC:** CloudFormation-deployed VPC and security.

OBJECTIVE

- **Resilience:** Implement Fault Tolerance using Dead Letter Queues (DLQ).
- **Performance:** Achieve <200ms API response time using Serverless.
- **Video Engineering:** Automate thumbnail generation and transcoding using FFmpeg on Linux.

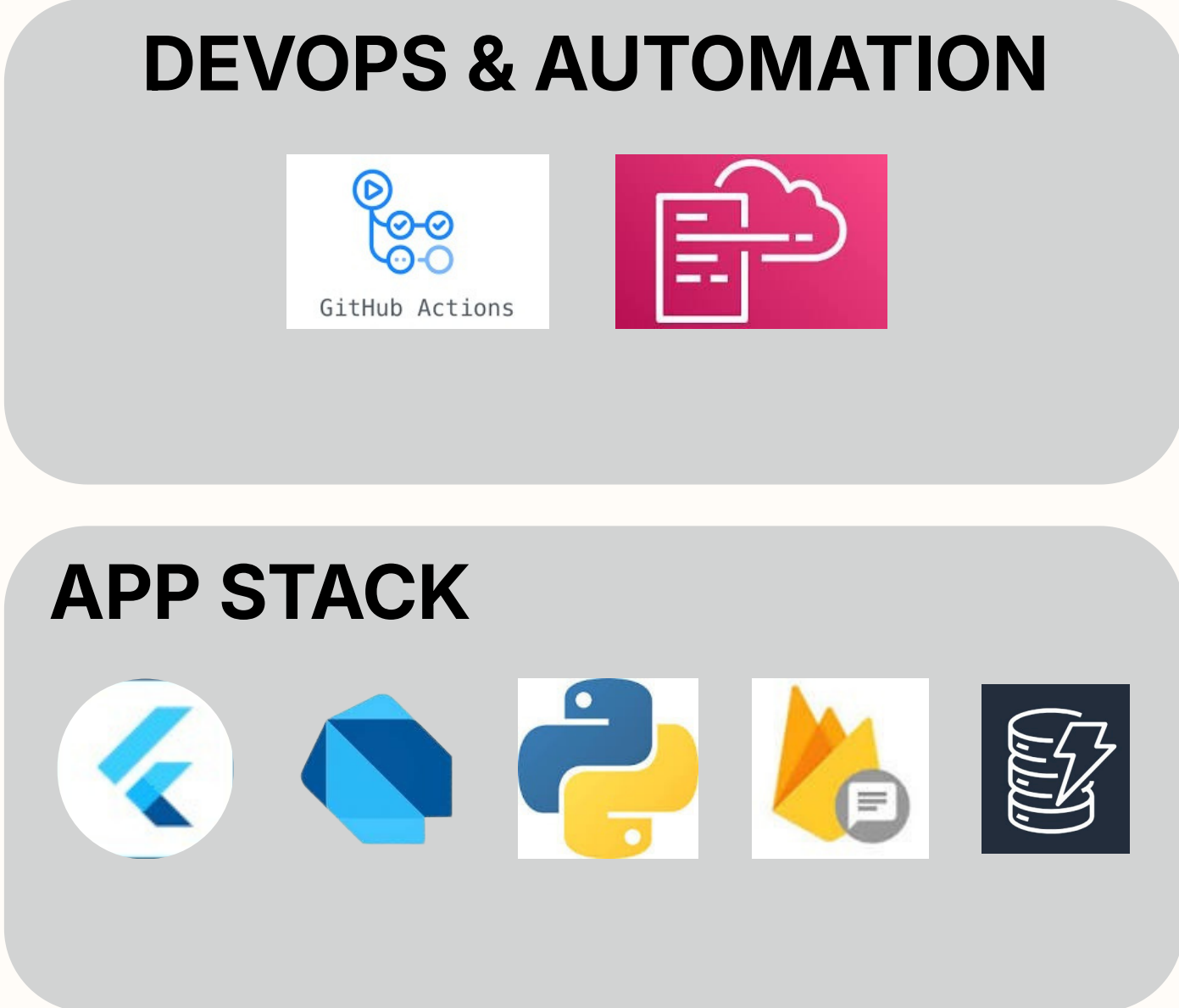
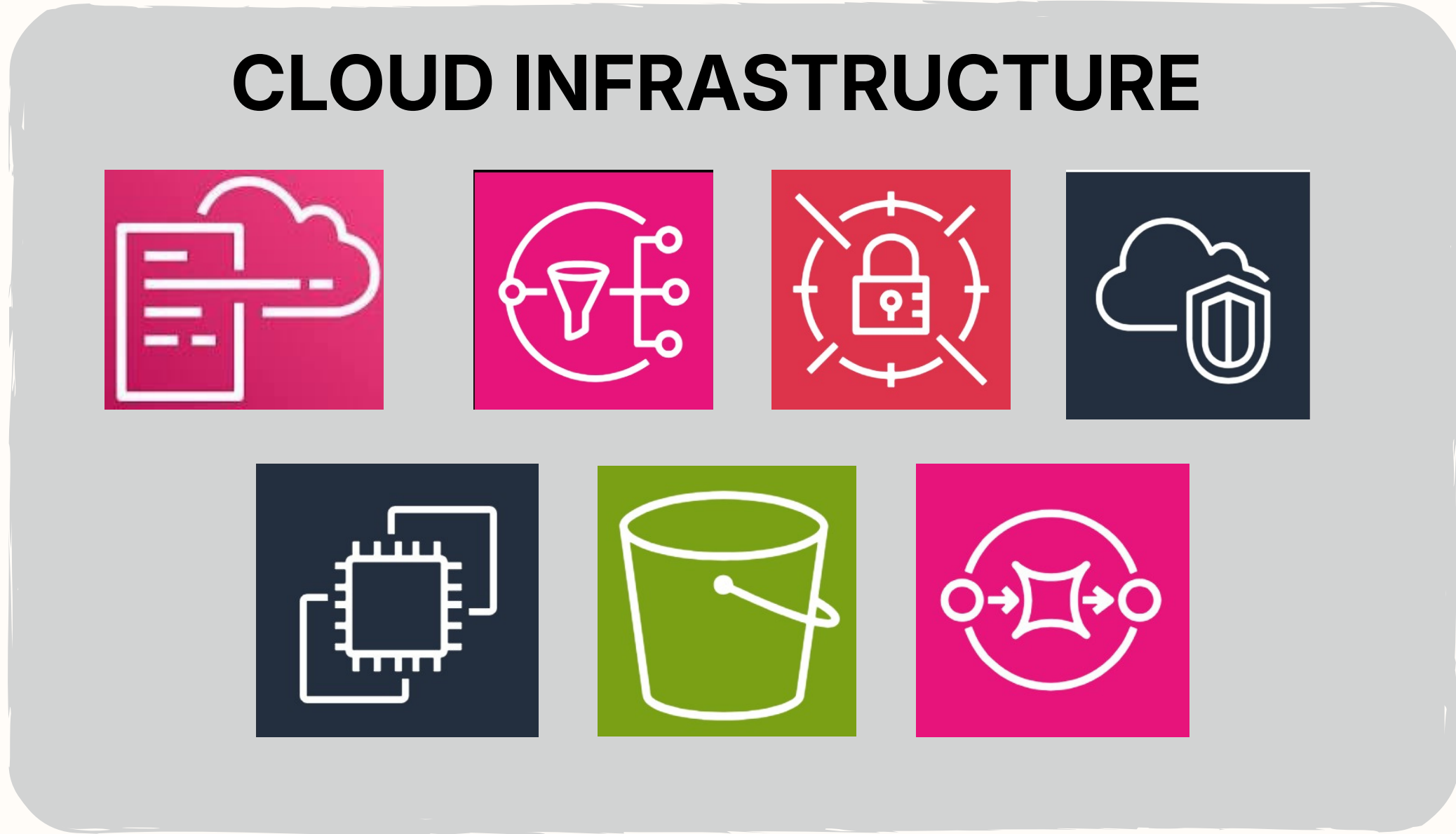
ANALYSIS & RESULTS

- **Enterprise Evolution:** Speed, Visibility, Security
- **Performance:** SQS offloading cut API latency by 85%.
- **Observability:** Real-time cost tracking via CloudWatch.
- **Security:** Zero-Trust VPC isolation and automated credential rotation.

CONCLUSION

EduStream bridges Serverless and EC2 to solve LMS scaling issues. With Firebase for instant notifications and a fully automated CI/CD pipeline, we've built a reactive, cost-effective video platform.

TOOLS AND TECH USED



REFERENCES

- **AWS Documentation:** Amazon SQS & Lambda Patterns.
- **Google Firebase:** Cloud Messaging for Flutter.
- **FFmpeg:** Digital Video Processing Documentation.
- **GitHub Repository:** [github.com/\[your-username\]/edustream](https://github.com/[your-username]/edustream)